Logic Gates · AS · Practice

U03a · 24 written + 6 pen-first = 30 marks · Name: _____ Due: _____

Questions labelled Ours are authored. Keep the supplied structure; use intermediate columns to show your reasoning.

1. Ours: follow the expression

Complete X = (NOT A) AND (B OR C). Working columns are optional. **[4]**

A	B	C	NOT A	B OR C	X
0	0	0			
0	0	1			
0	1	0			
0	1	1			
1	0	0			
1	0	1			
1	1	0			
1	1	1			

2. Ours: name the behaviour

Name (a) the gate that outputs 0 only for inputs 11, and (b) the gate that outputs 1 only for inputs 00. **[2]**

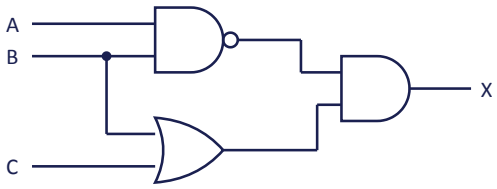
3. Ours: OR and XOR

State the two-input combination for which OR and XOR differ. Give both outputs. **[2]**

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4. Ours: read a circuit



(a) Write the logic expression for X. Preserve the structure of the circuit. **[3]**

(b) State what the small circle at the output of the upper gate means. **[1]**

5. Ours: equal functions?

A colleague proposes replacing the circuit with **Y = NOT (A AND B)**.

(a) Calculate X and Y when A=0, B=0 and C=0. **[2]**

(b) Use your result to decide whether X and Y are equivalent Boolean functions. Explain why this one input combination can settle the claim. **[2]**

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6. Ours: exam-shape rehearsal · trace the expression

A fresh authored circuit expression, using the complete-output-column question shape.

Complete X = (A NOR B) XOR C. **[2]**

A	B	C	Working space	X
0	0	0		
0	0	1		
0	1	0		
0	1	1		
1	0	0		
1	0	1		
1	1	0		
1	1	1		

7. Ours: a status light

A status light must be on when exactly one of two sensors A and B is 1.

(a) Write the expression for output L using one gate. **[2]**

(b) Complete the output column. **[2]**

A	B	L
0	0	
0	1	
1	0	
1	1	

(c) A colleague substitutes OR. Identify the input combination that reveals the error and give the incorrect and required outputs. **[2]**

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8. Ours: the mystery box · pen first [6]

💻 **Predict before touching the switches.** A two-input box has outputs **1, 1, 1, 0** for input rows **00, 01, 10, 11**. Write your answers first; then compare them with the deck's gate table or CMOS model. All six marks are human marked.

(a) Name the gate. State the input combination that gives output 0. **[2]**

(b) Both inputs are connected to the same signal A. Give the output when A=0 and when A=1. **[2]**

(c) Name the one-input function you have made. Explain why the two cases in (b) establish that function. **[2]**

Keyboard companion: Karel 09 HarvestTheField and 10 WallFollower remain the early-course programming work, outside these 30 marks. No Python file is required for this gate investigation.

Above and beyond · unmarked. Use only NAND gates to build AND. Explain why this construction does **not** prove that real chips contain only NAND gates.

Additional space - label the question number.